SuperBundle - Bundle Adjustment with Superquadric Primitives

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Abstract

Recovering accurate camera poses in sparse-view settings is hard: with few views, reprojection error alone leaves the cameras under-constrained and pose estimates drift. We study this in a known environment, where the scene geometry is available ahead of time. We add a compact superquadric scene prior to bundle adjustment, fit offline by SuperDec [3] from the ground truth point cloud and registered into the reconstruction frame with a similarity transform from known camera centres. The prior enters as a one-sided surface term. It pulls triangulated points that have drifted off the known surfaces back onto them, and leaves interior points free. Because the points move during optimization, we re-associate them to their nearest superquadric in an EM-style loop. We evaluate on ten scenes of the Aria Synthetic Environments Dataset [1]. Classic bundle adjustment accounts for most of the improvement. The prior adds a further gain in pose AUC@5, largest where views are scarce: up to $+1.5$ at six views. The gain shrinks to roughly $+0.1$ to $+0.2$ by eight to ten views, once enough views already fix the geometry. The prior helps where reprojection is weak.

1. Introduction

Reconstructing 3D scenes from multiple camera images is a classical problem in Computer Vision. There are many feed-forward models for predicting camera parameters and point clouds to reconstruct full 3D scenes [5, 12, 14]. However, especially in sparse view settings, camera pose estimates can drift or be imprecise. This comes from the limited overlap of the views, ambiguous geometry, and scale uncertainty. To reduce the prediction error one can use optimization based methods like bundle adjustment [11] to refine the reconstructed scenes.

In this project, we improved the camera poses in very sparse view settings with a modified version of the classic bundle adjustment algorithm. We used a superquadric [8] representation of our 3D scene as a structural prior, improving the camera pose estimates compared to the classic bundle adjustment. For this, we used the superquadric representation of the Aria Synthetic Environments Dataset[1] generated with Superdec [3] from the ground truth point clouds.

XXX Add stuff with connection to results and discussion

2. Related Works

Feed Forward Reconstruction Traditional multi-view reconstruction pipelines [4, 7, 9] rely on sequential stages of feature extraction, matching, and geometric optimization, making them computationally expensive and sensitive to initialization. Feed-forward reconstruction models address this by directly regressing 3D structure and camera parameters from images in a single forward pass. Feed-forward reconstruction models address this by directly regressing 3D structure and camera parameters from images in a single forward pass. DUSt3R [13]and MASt3R [6] established this paradigm by predicting dense point maps and relative camera poses from image pairs without requiring known intrinsics, relying on post-hoc global alignment to extend beyond pairs. Methods like VGGT [12] or MapAnything [5] advanced this further by processing up to hundreds of images simultaneously in a single forward pass, jointly predicting camera parameters, depth maps, point maps, and 3D point tracks.

Bundle Adjustment Bundle adjustment (BA) is the gold standard for refining camera poses and 3D structure in multi-view reconstruction. Formally introduced by Triggs et al. [11], it minimizes the total reprojection error over all cameras and scene points simultaneously via the Levenberg-Marquardt algorithm, exploiting the sparse block structure of the Jacobian to keep the problem tractable. In classical SfM pipelines such as COLMAP [9], BA is applied both locally after each image registration and globally as a final refinement, preventing drift and enforcing geometric consistency across the full reconstruction. While highly effective given sufficient image overlap and reliable correspondences, classical BA is sensitive to initialization and provides no mechanism to recover from fundamentally incorrect pose estimates. This limitation becomes particular acute in the sparse-view setting.

Superquadrics Superquadrics are a family of parametric geometric primitives, first introduced by Barr [2] and later popularized for 3D shape recovery by Solina and Bajcsy [10], capable of representing a wide range of shapes — from boxes to cylinders to ellipsoids — with a small set of parameters encoding position, orientation, scale, and two shape exponents. Their compactness and analytical differentiability make them attractive as structural scene representations. SuperDec [3] builds on this foundation by presenting an approach for creating compact 3D scene representations through decomposition into superquadric primitives, designing a novel architecture that efficiently decomposes point clouds of arbitrary objects into a compact set of superquadrics. Rather than targeting photorealistic reconstruction, SuperDec solves the decomposition locally on individual objects and leverages instance segmentation to scale to full 3D scenes. In our work, we use SuperDec to obtain a compact geometric abstraction of the scene from the ground truth point cloud, which we then use as a structural prior to constrain and guide bundle adjustment optimization.

3. Method

We are given a sparse set of N views of an indoor scene with known intrinsics and want accurate camera poses. The pipeline runs in two stages: a feed-forward stage that produces an initial reconstruction, and a bundle adjustment stage that refines the camera poses, with the superquadric prior entering as an extra cost in the second stage.

Initialization and correspondences We initialize the cameras and scene geometry with VGGT [12] through the MapAnything harness [5], which gives a camera pose and a dense depth map per view in a single forward pass. To get the 2D–2D matches that bundle adjustment optimizes against, we run MASt3R [6] on every ordered view pair and keep its dense reciprocal correspondences above a confidence threshold. Each surviving match is unprojected to a camera-frame ray with the VGGT intrinsics, rotated into the world frame using the VGGT poses, and triangulated by closest-approach midpoint of the two rays. We discard near-parallel rays, and we keep a triangulated point only if it has positive depth in both cameras and reprojects inside both images. This yields the initial 3D points X_j and their observations x_{ij}.

Bundle adjustment objective Each camera i is parameterized by a world-to-camera rotation R_i and translation t_i; the intrinsics are held fixed during optimization to avoid the focal-length/depth ambiguity. The base cost is the robustified reprojection error over all observations,

$$E_{\text{reproj}} = \sum_{i,j} \rho\big(\|\pi(R_i, t_i, X_j) - x_{ij}\|^2\big)\,, \tag{1}$$

where π is the pinhole projection and ρ is the Huber loss with threshold δ (default 2.0 px, set to 1.0 px in our reported runs), which down-weights gross MASt3R mismatches. We solve this least-squares problem with Ceres using a sparse Schur factorization, optimizing cameras and points jointly. The gauge is fixed by holding the first camera constant, which removes the 6-DoF freedom of the global frame.

Superquadric surface prior Reprojection alone underconstrains the cameras when views barely overlap. We add a structural prior that pulls the reconstructed points onto the known scene geometry, expressed as a set of superquadrics [3, 8] fit offline (see below). Each point X_j is associated to one superquadric $s(j)$ and mapped into that primitive's canonical frame, $q_j = R_{s(j)}^{\top}(X_j - t_{s(j)})$, where R_s and t_s are the primitive's rotation and centre. The inside–outside function of a superquadric with half-extents a_1, a_2, a_3 and shape exponents $\varepsilon_1, \varepsilon_2$ is

$$F(q) = \left(\left|\frac{q_x}{a_1}\right|^{2/\varepsilon_2} + \left|\frac{q_y}{a_2}\right|^{2/\varepsilon_2}\right)^{\varepsilon_2/\varepsilon_1} + \left|\frac{q_z}{a_3}\right|^{2/\varepsilon_1}\,, \tag{2}$$

in the inverse Solina form [10], with $F^{-\varepsilon_1/2} - 1$ positive inside the surface, negative outside, and zero exactly on it. The full objective adds a weighted one-sided surface term to the reprojection cost:

$$E = E_{\text{reproj}} + \sum_j \Big(\lambda\,\|q_j\|\,\max\big(0,\,1 - F(q_j)^{-\varepsilon_1/2}\big)\Big)^2. \tag{3}$$

The $\max(0, \cdot)$ makes the term a hinge: it penalizes only points that lie *outside* their associated surface and leaves interior points free, so the prior pulls points that have drifted off the scene back onto it and leaves points already inside a primitive alone. The factor $\|q_j\|$ turns the unitless inside–outside value into an approximate radial distance in meters. The weight λ is a pixels-per-meter factor that puts the surface residual on the same scale as the reprojection residual; setting $\lambda = 0$ recovers plain reprojection BA. We use $\lambda = 15$ and a separate Huber of 2.749 on the surface term.

Both costs are summed into one Ceres problem and minimized together.

Association and EM refinement Each point is bound to its nearest superquadric by radial distance, $r(p) = \|q\|\,|F(q)^{-\varepsilon_1/2} - 1|$, taking the argmin over the active primitives. A point whose nearest-primitive distance exceeds a cutoff (we use 0.0372 m) is left unassigned and contributes no surface term, which keeps stray triangulations from being pulled toward unrelated geometry. Because Ceres moves the points during optimization, a fixed association computed at the start drifts out of date. We therefore re-estimate it in an EM-style loop: a reprojection-only warm-up solve first cleans up the structure so the first association is reliable, then we alternate an E-step that re-associates the current points with an M-step that runs a short surface BA solve. We use two outer iterations of 41 inner Ceres iterations each. With one outer iteration this reduces to associate-once, solve-once.

Prior registration The superquadrics are fit offline by SuperDec [3] on the clean, full-scene point cloud back-projected from ground-truth depth and instance masks, and they live in the scene's world frame. Our setting is camera-pose refinement in a *known* environment, so this geometry is available as a fixed prior rather than test-time information about the cameras we solve for. To use it, we bring the superquadrics from the scene frame into the reconstruction frame produced by VGGT. We fit a $\mathrm{Sim}(3)$ from the ground-truth camera centres to the predicted camera centres with Umeyama alignment, invert it, and apply the inverse to the primitives before association. This step uses ground-truth camera centres only to register the prior; the camera *poses* are still what bundle adjustment refines.

4. Results

Experimental setup We evaluate on the Aria Synthetic Environments (ASE) dataset [1], using ten scenes. For each scene we select sparse view subsets with a covisibility criterion: we precompute a pairwise covisibility matrix from the ground truth depth and keep view sets whose pairwise covisibility meets a threshold of 0.6. We sweep the number of input views over 4, 6, 8, and 10 to study how the prior behaves as views become scarcer. The seed is fixed and view selection is deterministic, so the same view sets feed every configuration, and only the post-reconstruction optimization differs.

The superquadric prior is fit offline by SuperDec [3] on the clean full-scene ground truth point cloud, a known-environment assumption. To use it during optimization we register the world-frame superquadrics into the predicted reconstruction frame with a Sim(3) estimated from ground truth camera centres. This is an idealized known-map setting, not full relocalization: ground truth poses enter the pipeline only to place the prior, and the optimization never sees them as targets.

	Pose AUC@5 ↑ (# views)			
Configuration	4	6	8	10
VGGT-only	27.7	20.4	13.4	9.3
BA baseline	39.3	29.6	27.9	29.4†
Ours (BA + prior)	**40.7**	**31.1**	**28.0**	**29.6**
Gain from prior	+1.3	+1.5	+0.1	+0.2

Table 1. Pose AUC@5 (↑, scaled to 0–100, averaged over the ten ASE scenes) by input view count. VGGT-only is the raw feed-forward output; the BA baseline is reprojection-only bundle adjustment ($\lambda = 0$); Ours adds the superquadric surface prior. The last row is the prior's gain over the baseline. The prior helps most when views are few. At 8 and 10 views the baseline already pins the geometry, so the gain is near zero. †The 10-view baseline is our regular reprojection-BA result on the MASt3R backend; we do not have a matched $\lambda = 0$ run at 10 views, so the $+0.2$ there compares against this bar rather than an exactly matched baseline.

We report pose AUC@5, the area under the relative-pose accuracy curve up to a $5°$ threshold. For every unordered view pair we measure the rotation-angle and translation-direction-angle error between the predicted and ground truth relative poses, take the larger of the two, and accumulate the fraction of pairs below each integer-degree threshold up to $5°$. The area under this curve is scaled to 0–100; higher is better. Per-sample values are pooled across all ten scenes and averaged.

We compare three configurations on these fixed view sets. VGGT-only is the raw feed-forward output with no optimization. The BA baseline runs our bundle adjustment with the surface term switched off ($\lambda = 0$), leaving pure Huber reprojection. Ours adds the superquadric surface prior on top of the same reprojection cost. All bundle adjustment runs in Ceres; cameras are initialized from VGGT through the MapAnything harness [5, 12], and 2D–2D correspondences come from MASt3R [6], midpoint-triangulated into the world frame before optimization.

Main results Table 1 reports pose AUC@5 for the three configurations across the four view counts. Bundle adjustment accounts for almost all of the improvement over raw VGGT: at 10 views it lifts pose AUC@5 from 9.3 to 29.4, and at 4 views from 27.7 to 39.3. The superquadric prior adds a further gain on top of the baseline that is largest in the sparsest settings, $+1.3$ at 4 views and $+1.5$ at 6 views, and shrinks to $+0.1$ at 8 views and $+0.2$ at 10 views.

Analysis The prior helps where reprojection alone leaves the cameras under-constrained, and tapers off once enough views fix the geometry. With many overlapping views the reprojection term already determines the camera poses, so the surface residual has little left to correct. The sparse case is different. With four or six low-covisibility views the matched structure is thin and the reprojection term is weak. The prior then pulls triangulated points back onto the known scene surfaces, supplying the constraint that is missing.

Sweeping the surface weight per view count tells the same story. At four views the fixed-weight prior gives $+1.3$, but the per-view sweep reaches $+2.0$ at $\lambda = 100$ (baseline 39.3 versus 41.3); the sweep is noisy and non-monotonic, with $\lambda = 60$ falling back to the baseline and $\lambda = 200$ dropping again to 40.3. At six views the prior helps 3 of the 10 scenes and hurts none: scene 6 improves by $+9.3$ ($48.0 \rightarrow 57.3$), scene 2 by $+4.0$, and scene 9 by $+1.3$, while the other seven scenes are unchanged. So the aggregate six-view gain comes from three scenes, not a shift across all ten. A structural prior only binds where the local geometry is poorly conditioned, so a few badly-conditioned scenes carrying the aggregate gain is the expected behaviour.

5. Discussion

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